10. Find the degree of $\left(2a^2+3ab-b^2\right)-\left(3a^2-ab-3b^2\right)$

11. Find the value of w, given that $x=3$, $y=4$, $z=-2$ and $w=x^2-y^2+z^2-xyz$.

12. Simplify and find the degree of $6x^2+1-\left[8x-\left\{3x^2-7-\left(4x^2-2x+5x+9\right)\right\}\right]$

13. The two adjacent sides of a rectangle are $2x^2-5xy+3z^2$ and $4xy-x^2-z^2$. Find the perimeter and the degree of the expression.

DO YOU KNOW?

Srinivasa Ramanujan, the great Indian mathematician created so many beautiful equations in his childhood using exponents. Here is an interesting exponential equation from his most popular "**Notebooks**".



$$2^2\times6^6\times1^1\times1^1=3^3\times3^3\times4^4$$

The base and power are equal in each factor. Moreover the sum of base (or power) in both the sides are also equal. (that is, $2+6+1+1=3+3+4=10$). It can be easily proved with the help of laws of exponents.

$$\text{LHS}=2^2\times6^6\times1^1\times1^1=2^2\times6^6\times1=2^2\times(2\times3)^6$$

$$=2^2\times2^6\times3^6 \qquad [\text{since, } (a\times b)^m=a^m\times b^m]$$

$$=2^{2+6}\times3^{3+3} \qquad [\text{since, } a^m\times a^n=a^{m+n}]$$

$$=2^8\times3^3\times3^3$$

$$=2^{2\times4}\times3^3\times3^3$$

$$=\left(2^2\right)^4\times3^3\times3^3 \qquad [\text{since, } a^{m\times n}=\left(a^m\right)^n]$$

$$=4^4\times3^3\times3^3=3^3\times3^3\times4^4$$

$$=\text{RHS.}$$

Try to prove some of his other equations given below:

$$8^8\times9^9\times1^1=3^3\times3^3\times12^{12} \quad \text{(Base total is 18)}$$

$$4^4\times20^{20}\times30^{30}\times1^1=6^6\times24^{24}\times25^{25} \quad \text{(Base total is 55)}$$

Summary

- If 'a' is any integer, then $a\times a\times a\times\ldots\times a$ (n times) $=a^n$. Here 'a' is the base and n is the exponent or power or index.
- $(-1)^n=\begin{cases}1, & \text{if } n \text{ is even}\\ -1, & \text{if } n \text{ is odd}\end{cases}$
- When a number 'a' is multiplied by itself, the product is called the square of that number and denoted by a^2. Similarly, the square of a number (a^2) is multiplied by 'a', then the product is called the cube of that number and is denoted by a^3.

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